AICE2005 Integrated Design Assignment

Designing and Vibe-Coding a Note-Taking Application

Overview

Across the workshop series, your team designs a note-taking application from first principles — personas, stakeholders, requirements, UML design, prioritisation, and a test plan — then uses that specification to **vibe code** a working implementation with an AI coding assistant. The assignment brings these two halves together: a **specification document** built from your workshop artefacts, and a **working application** built from that specification, defended in a **group presentation with individual oral questioning**.

Learning Outcomes

By completing this assignment, you will demonstrate that you can:

- Identify personas and connect user needs to application behaviour (W1).
- Map stakeholders by power and interest, and translate that into functional and non-functional requirements (W2).
- Express a system's structure and behaviour using UML class and sequence diagrams (W3).
- Prioritise and estimate implementation work using MoSCoW or T-shirt sizing (W4).
- Refine your UML into a design-level diagram using standard notation — visibility, types, multiplicity, relationships and packages (W5).
- Prioritise your requirements specification using MoSCoW, with each priority justified against personas, stakeholders or use cases (W6).
- Plan how your requirements will be tested, with a test case for each requirement (W7).
- Use a specification to direct an AI-assisted ("vibe coded") implementation, and explain where the implementation does — or doesn't — match what was specified.
- Individually explain and defend your team's design and implementation decisions under direct questioning.

Deliverables

Each group (up to five members) submits two components: a **specification document** and a **working application**, presented together in a **group demo with individual oral questioning**.

1) Specification Document (PDF)

Compiled from the artefacts produced across the workshops. Carry each template's content into the final document rather than submitting the raw templates:

Section	Source workshop
1. Personas & Use Case Diagram	W1
2. Stakeholder Mapping & Functional / Non-Functional Requirements	W2
3. UML Class & Sequence Diagrams	W3
4. Task Prioritisation — MoSCoW / T-shirt Sizing	W4
5. UML Design Diagram	W5
6. Requirements Prioritisation — MoSCoW	W6
7. Test Plan	W7

A team logo (W0) may be used as a cover page, optional.

2) Vibe-Coded Application

A working note-taking application, built by directing an AI coding assistant using your specification. Submit a link to the repository (with commit history visible) alongside the specification document.

3) Group Presentation & Individual Oral Q&A

A short group demo of the working application against your specification, followed by individual questioning — each team member is asked questions directly about the design and implementation, to assess their individual understanding of and contribution to the work.

Recommended structure:

- Slide 1: Introduce your app and your team.
- Slide 2: Personas & the problem you're solving (W1).
- Slide 3: Requirements, their MoSCoW priorities & how they shaped your design (W2, W6).
- Slide 4: System design — class/sequence and design diagrams (W3, W5).
- Slide 5: What you built vs. what you specified — and why they diverge, if they do.
- Live demo of the application.
- Individual oral questions (no slide — asked directly during/after the demo).

Submission Details

- Due: 30/11/26 (via Moodle).
- Presentation: Week 11 and 12 workshop session.
- Format: Group PDF specification + repository link + live demo.
- Weight: 40% of module grade.

Assessment

Marks are split across the specification document, the application, and the presentation/individual questioning. Suggested weighting (confirm/adjust before publishing):

Component	Weight
Specification document	60
Vibe-coded application	20
Presentation & individual oral Q&A	20

Specification Document Rubric (60 marks)

Criterion	Weight	Excellent	Good	Satisfactory	Poor
Personas & Use Case Diagram	10	**(8–10)** Personas detailed and linked to clear, consistent use cases.	**(6–7)** Mostly clear; minor gaps in linkage.	**(4–5)** Generic personas or weak use case linkage.	**(0–3)** Missing or poor quality.
Stakeholder Mapping	8	**(7–8)** All quadrants populated with justified, specific stakeholders.	**(5–6)** Mostly complete; light justification.	**(4)** Sparse or generic stakeholders.	**(0–3)** Missing.
Functional Requirements	10	**(8–10)** Requirements precise, traceable to personas/use cases.	**(6–7)** Mostly clear; some untraceable.	**(4–5)** Vague or incomplete.	**(0–3)** Missing.
Non-Functional Requirements	8	**(7–8)** Specific, measurable, relevant to the app.	**(5–6)** Mostly reasonable.	**(4)** Generic or unmeasurable.	**(0–3)** Missing.
UML Design (Class, Sequence & Design Diagram)	12	**(10–12)** Diagrams consistent with requirements; relationships and flows clear; design diagram uses correct notation (visibility, types, multiplicity, packages).	**(8–9)** Mostly coherent; minor inconsistencies.	**(5–7)** Partial or inconsistent with requirements.	**(0–4)** Missing or incorrect.
Prioritisation (Requirements MoSCoW & Task MoSCoW / T-shirt)	6	**(5–6)** Requirement priorities and task priorities/sizes justified, balanced and realistic.	**(4)** Mostly reasonable; light justification.	**(3)** Uneven or arbitrary.	**(0–2)** Missing.
Test Plan	6	**(5–6)** All requirements covered by clear, measurable test cases.	**(4)** Mostly mapped.	**(3)** Partial coverage.	**(0–2)** Missing.

Application & Presentation Rubric (40 marks)

Criterion	Weight	Excellent	Good	Satisfactory	Poor
Specification-to-implementation fidelity	12	**(10–12)** App clearly implements the specified requirements; any deviations are explained and justified.	**(8–9)** Mostly matches spec; minor unexplained gaps.	**(5–7)** Partial match; gaps not addressed.	**(0–4)** App bears little relation to the spec.
Application quality & functionality	8	**(7–8)** Application is functional, usable, and demonstrates the core note-taking flows.	**(5–6)** Mostly functional; minor issues.	**(4)** Partially functional.	**(0–3)** Non-functional or incomplete.
Group demo	10	**(8–10)** Confident, well-paced demo that clearly shows the app working end to end.	**(6–7)** Mostly clear; minor stumbles.	**(4–5)** Demo is unclear or incomplete.	**(0–3)** No working demo.
Individual oral Q&A	10 (per student)	**(8–10)** Clearly explains and defends their own contribution and the team's design decisions.	**(6–7)** Mostly able to explain decisions, with some prompting.	**(4–5)** Limited understanding of decisions beyond their own part.	**(0–3)** Unable to explain the design or their contribution.

Individual oral Q&A is scored per student, independently of the group marks above — it is how each team member's individual understanding and contribution are assessed.